Analyzing a Real-World Dataset Using Linear Regression

Introduction

This project uses linear regression to investigate the relationship between socio-economic and law-enforcement variables and violent crime in U.S. communities, using the UCI Communities and Crime dataset. There are 1,994 observations and 127 potential predictors, with the response ViolentCrimesPerPop, a standardized per-capita crime rate. The data include a mix of census, policing, housing, demographic and social structure variables and as such are well suited for an applied regression project with the focus on both prediction and interpretation.

Not just to fit a model with good predictive power, but also to construct an interpretable regression specification to facilitate the interpretation of coefficients, hypothesis testing, the construction of the confidence interval, and formal model diagnostics is the primary goal of this analysis. For that reason, the final workflow combines data cleaning, penalized least-squares screening, ordinary least squares estimation, model comparison, and diagnostic testing. The resulting model demonstrates that violent crime has a close relationship with family structure, housing distress, population density, and homelessness-related variables, and many law-enforcement variables are less strongly related to violent crime due to the large amount of missing data.

One of the major warnings is that it is an observational study. Thus, any estimated relationships must be viewed as conditional relationships but not causal influences. This is particularly necessary when some demographic variables are sensitive, as this might indicate latent structural inequality, segregation or omitted contextual variables as opposed to direct causal processes.

Data Preprocessing

Data cleaning was the initial preprocessing operation. Communities and Crime dataset are already numerically normalized by attribute in the 0-1 range. This implies that no further min-max normalization had to be done on the OLS step, since all predictors are already represented by numbers of a similar scale by their own attributes. Nevertheless, it is also necessary to keep in mind that each of the variables was normalized separately. Due to this fact, coefficients can be interpreted directionally and within-variable movement, but a comparison of the magnitude of coefficients across variables should be done carefully.

Five non-predictive columns were removed prior to modeling: state, county, community, communityname and fold. These variables are used to identify the observation, as metadata, or to do cross-validation bookkeeping. Specifically, the fold variable must not be adopted as a predictor at all since it is only there to specify the given 10-fold cross-validation split.

Missing data was the significant cleaning issue. The missingness of many of the policing and LEMAS variables was extremely high. Variables with a drop rate exceeding 80% were dropped altogether. This dropped 22 columns, such as staffing, budget, patrol, overtime, police-race composition and request-load measures. When these very incomplete variables and the five non-predictive fields were eliminated, there were 100 predictors left. There was then only a single cell missing in the retained

feature matrix, which was imputed with the median. Median imputation was chosen over the others since it is resistant to skew and outliers and is a safe method of normalized quantitative variables.

Though, no additional normalization was necessary when using OLS, standardization was applied in the penalized least-squares step. This is significant since lasso and elastic net penalties are scale sensitive; otherwise variables of greater variance may overrule the shrinkage process. Thus, feature screening was done using standardized predictors and the final OLS model was estimated on the original normalized predictors to be interpreted.

Model Definition and Classical Linear Model Assumptions

The final explanatory model is a multiple linear regression of the form

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_{13} X_{13i} + u_i,$$

where Y_i is the normalized violent crime measure for community i, and the selected predictors are:

racepctblack, agePct12t29, pctUrban, OtherPerCap, MalePctDivorce, PctKids2Par, PctWorkMom, PctIlleg, PctPersDenseHous, HousVacant, PctHousOccup, MedOwnCostPctIncNoMtg, and NumStreet.

The classical linear model assumptions underlying OLS are:

1. Linearity in parameters: the conditional mean of the outcome can be represented as a linear function of the included regressors and coefficients.
2. Random sampling / independence of observations: each community contributes one independent observation.
3. No perfect multicollinearity: no predictor is an exact linear combination of the others.
4. Zero conditional mean: the error term has expected value zero given the included regressors.
5. Homoskedasticity: the variance of the error term is constant across observations.
6. Normality of errors: mainly needed for exact small-sample inference.
7. No autocorrelation: errors are uncorrelated across observations.

In cross-sectional community data, the most realistic threats are omitted variables, heteroskedasticity, non-normal residuals, and multicollinearity among socially related indicators. These are therefore examined explicitly in the diagnostics section.

Estimation of Model Parameters

The estimation process involved two steps. A BIC-based lasso screening step was first used to narrow down the 100 filtered predictors to 20 variables. Second, BIC-based backward elimination created a more understandable 13 variables OLS specification. This combination approaches prediction and interpretability: the least squares with penalty is used to deal with high dimensionality, and the resulting OLS model retains the usual way of doing t-tests, F-tests, and confidence intervals.

Variable	**Coefficient**	**Effect per +0.10**	**HC3 SE**	**HC3 p-value**	**95% HC3 CI**
const	0.3535	0.0353	0.0543	0.0000	[0.247, 0.460]
racepctblack	0.1695	0.0169	0.0292	0.0000	[0.112, 0.227]
agePct12t29	-0.0917	-0.0092	0.0226	0.0000	[-0.136, -0.047]

pctUrban	0.0378	0.0038	0.0080	0.0000	[0.022, 0.053]
OtherPerCap	0.0482	0.0048	0.0164	0.0033	[0.016, 0.080]
MalePctDivorce	0.1097	0.0110	0.0306	0.0003	[0.050, 0.170]
PctKids2Par	-0.2953	-0.0295	0.0485	0.0000	[-0.391, -0.200]
PctWorkMom	-0.0794	-0.0079	0.0200	0.0001	[-0.119, -0.040]
PctIlleg	0.2168	0.0217	0.0458	0.0000	[0.127, 0.307]
PctPersDenseHous	0.1941	0.0194	0.0233	0.0000	[0.148, 0.240]
HousVacant	0.1113	0.0111	0.0381	0.0036	[0.037, 0.186]
PctHousOccup	-0.0637	-0.0064	0.0209	0.0024	[-0.105, -0.023]
MedOwnCostPctIncNoMtg	-0.0578	-0.0058	0.0164	0.0004	[-0.090, -0.026]
NumStreet	0.1808	0.0181	0.0539	0.0008	[0.075, 0.286]

The final 13-variable OLS model achieved an adjusted R^2 of 0.6619, a cross-validated RMSE of 0.1362, a cross-validated MAE of 0.0944, and a cross-validated R^2 of 0.6579. These values indicate that the model explains a substantial portion of the variation in the normalized violent-crime measure while still generalizing reasonably well in 10-fold cross-validation.

The estimated coefficients were interpreted with HC3 robust standard errors as heteroskedasticity was determined later on. There are a number of coefficients that are substantively significant. The coefficient of PctKids2Par is approximately equal to -0.2953, which is the greatest absolute impact in the latter model. With all other regressors held constant, the normalized share of children living with two parents is positively related to the normalized violent-crime rate. Since the predictor is a 0 to 1 scaled variable, a 0.10 value increases of this predictor will result in a predicted change of approximately 0.0295 in the normalized outcome.

The PctIlleg coefficient is positive at 0.2168 meaning that, communities with high proportions of children born to never-married mothers are more likely to have higher rates of violent-crime even after adjusting the other variables. PctPersDenseHous is highly positive as well at 0.1941 so that there is positive relation between dense housing and violent crime. NumStreet, a homelessness measure in the street, has a coefficient of 0.1808, once again exhibiting a positive correlation with crime. MalePctDivorce is positive at 0.1097 whereas, PctWorkMom is negative at -0.0794. HousVacant is positive and PctHousOccup is negative combined with the implication that housing instability and vacancy correlates with increased crime.

The coefficient on racepctblack is positive at 0.1695 though it should be taken with caution. Such a coefficient is not to be understood causally in an observational dataset like this one, or as an inherent trait. It more likely captures correlated structural factors such as historical segregation, concentrated disadvantage, or omitted local conditions. The negative agePct12t29 coefficient also speaks indirectly; when most family, density, and housing factors are held constant, the residual part of the relationship might not be susceptible to a one-to-one explanation.

Model Selection

Several competing models were compared. The full OLS model using all 100 cleaned predictors had the highest adjusted R^2 at 0.6785 and the lowest AIC at -2315.29, but it also had a very high condition number of 916.40, which signals serious multicollinearity and weak interpretability. The 20-variable post-lasso OLS model had adjusted R^2=0.6642, RMSE = 0.1361, and CV R^2=0.6588. The final 13-variable OLS model had R^2=0.6619, RMSE = 0.1362, MAE = 0.0944, and CV R^2=0.6579. Although its in-sample fit was slightly below the larger models, the drop in predictive performance was negligible. At the same time, the final model achieved the best BIC (-2220.95), which explicitly rewards parsimony.

Model	p	Adj_R2	AIC	BIC	RMSE	MAE	CV_R2
OLS_all100	100	0.678502861	-2315.290646	-1749.902953	0.136309602	0.095952134	0.657535786
OLS_postLasso20	20	0.6641512	-2305.671496	-2188.115639	0.13605185	0.09456772	0.658829711
OLS_final13	13	0.661905393	-2299.3201	-2220.949529	0.136229547	0.094419018	0.657937929
ElasticNet100	20				0.135590004	0.094245678	0.661142074

An Elastic Net model was also estimated as a pure prediction benchmark. It achieved the lowest cross-validated RMSE (0.1356) and the highest CV R^2 (0.6611), which makes it the best predictive model among the candidates. However, penalized models shrink coefficients and complicate classical coefficient-level inference. Since the main goal of the project is explanatory linear regression rather than only prediction, the final 13-variable OLS model is the most appropriate choice.

The final model is therefore justified on three grounds: it is parsimonious, it has nearly identical out-of-sample accuracy to the larger models, and it supports transparent inference with standard regression tools.

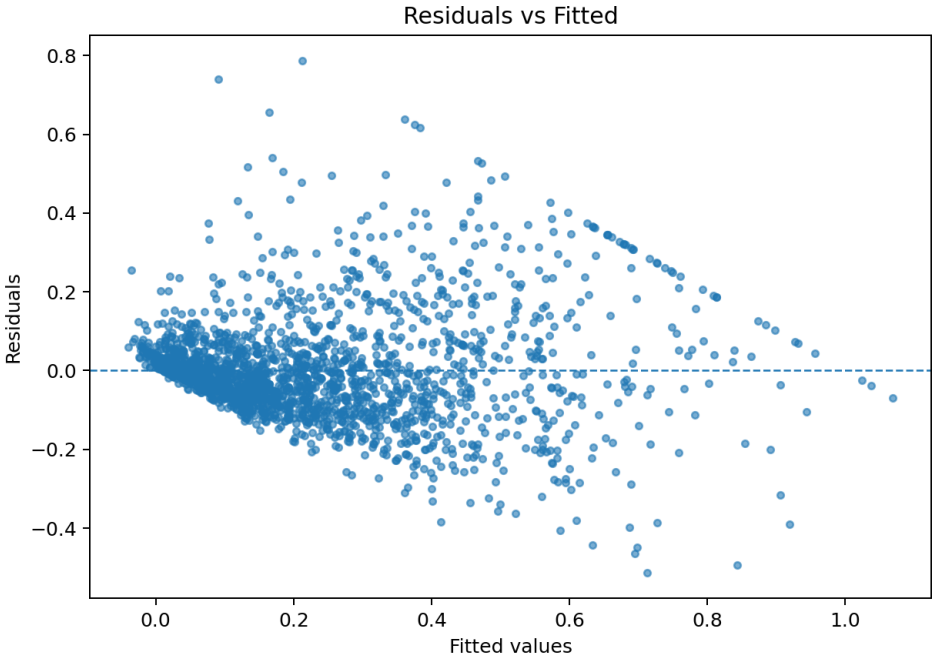
Model Diagnostics and Remedies

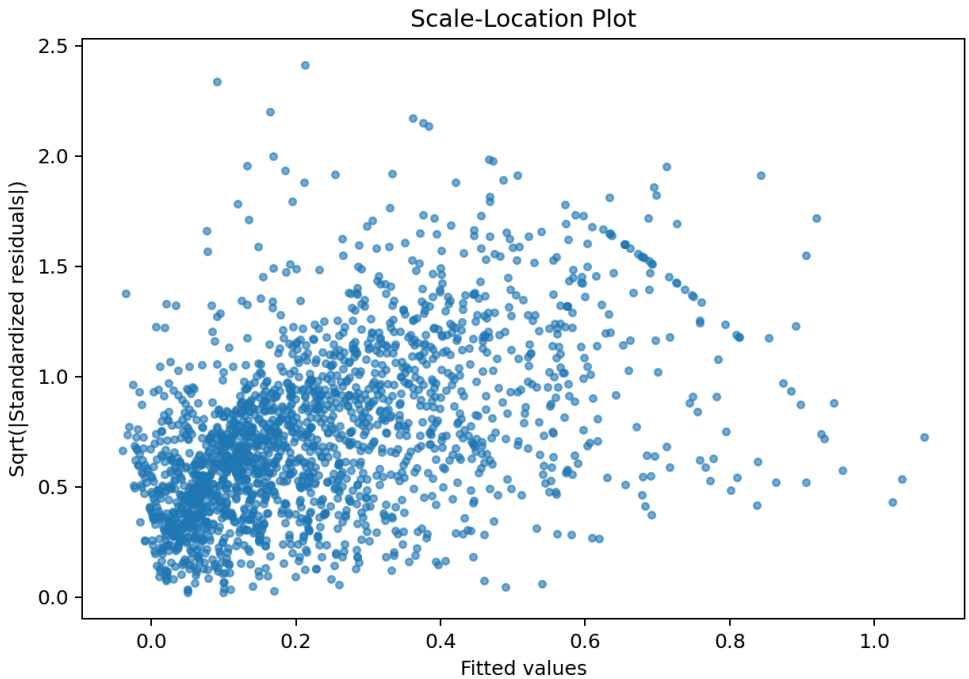
The diagnostics reveal that the final model is useful but not perfect.

variable	VIF
PctKids2Par	9.12080329
PctIlleg	7.938589308
racepctblack	3.646689687
MalePctDivorce	2.882174755
HousVacant	2.114960835
PctPersDenseHous	1.774774385
NumStreet	1.678013572
PctHousOccup	1.590597405
agePct12t29	1.330372616
PctWorkMom	1.302817807
MedOwnCostPctIncNoMtg	1.238740232

pctUrban	1.202679856
OtherPerCap	1.153704843

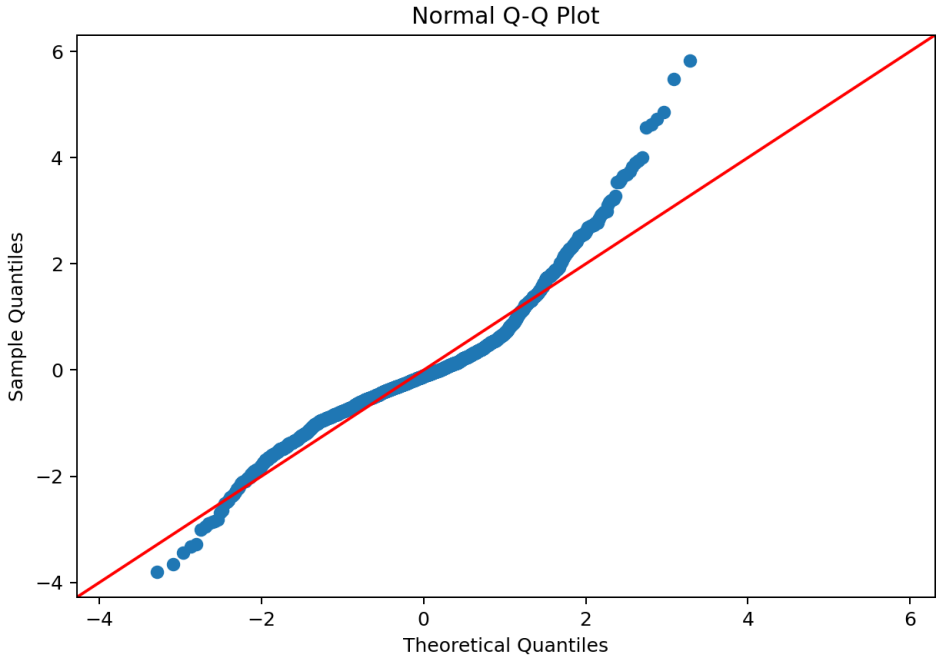
First, multicollinearity was significantly enhanced as compared to the complete model. The condition number of the final model is 42.93, whereas the full OLS has a condition number of 916.40. The VIF table indicates that most predictors have low to moderate VIFs with the highest value of PctKids2Par (9.12) and PctIlleg (7.94). These are not inconsequential, yet they are elucidable since both variables represent overlapping aspects of family structure. Variable screening and BIC-based pruning was the primary remedy applied here and it led to a significant decrease in instability.





Second, heteroskedasticity is clearly present. The Breusch–Pagan test strongly rejects homoskedasticity, with an F-test p-value around 2.15×10^{-48}. This is also visible in the Residuals vs

Fitted and Scale-Location plots, which show a fan-shaped pattern where residual spread increases with fitted values. To address this, HC3 robust standard errors were used for the reported inference. This is an appropriate and implemented remedy because it makes standard errors, t-statistics, p-values, and confidence intervals more reliable under heteroskedasticity.

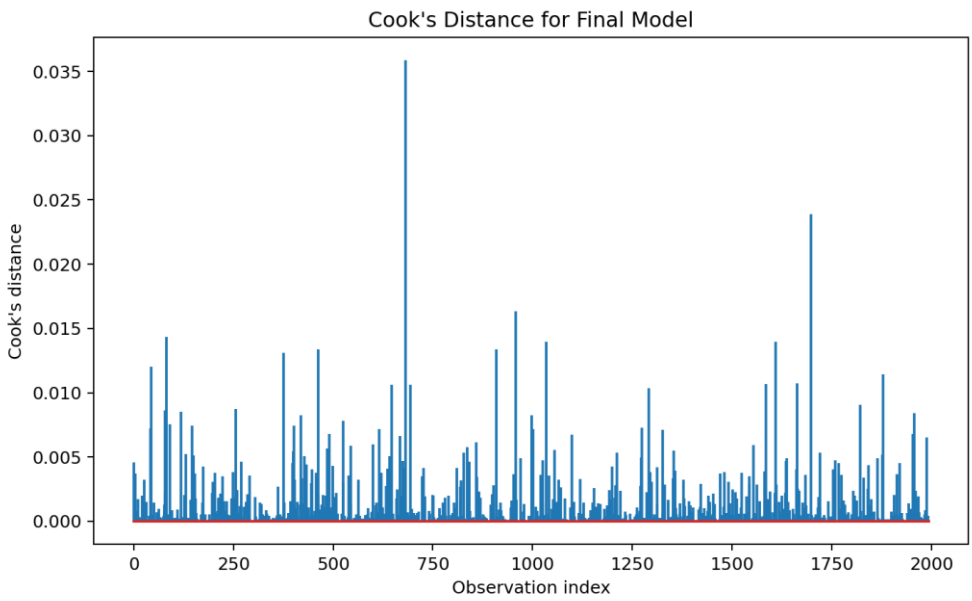


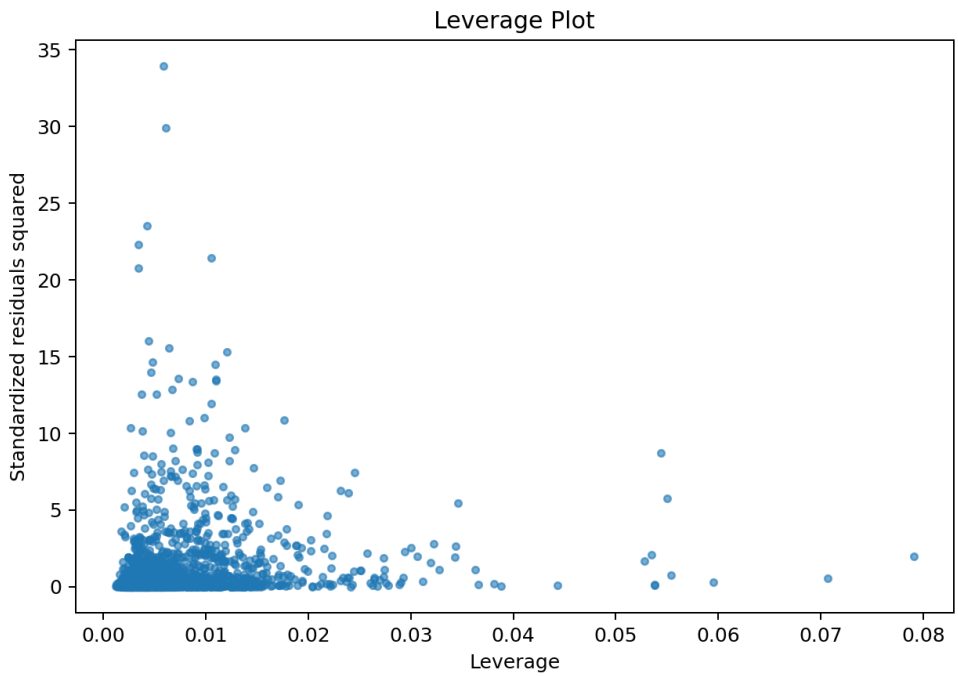
Third, the residuals do not follow the normal distribution. Jarque-Braun statistic is 1351.46 with p-value being almost 0 and the normal Q-Q plot illustrates that there is a strong upper-tail deviation toward the 45-degree line. This implies that the residual distribution is skewed to the right and heavy-tailed. Having a large sample of 1,994, this issue is not as harmful to asymptotic inference as it would be with a small sample, though it raises an alarm. Part of this problem is addressed by robust inference, but an extension in the future might investigate transformed or nonlinear models.

Fourth, the Ramsey RESET test does not accept the null of a correct functional form (F=13.01, p=0.00032). This implies that there could be nonlinearities or interactions that could be missing in the actual relationship. This is in line with the residual plots. One future enhancement would be to include quadratic terms or spline terms of the most significant continuous predictors like the PctKids2Par, PctPersDenseHous and NumStreet. With that said, in the case of a project based on classical linear regression, the existing model is an informative baseline.

Fifth, there is no evidence of meaningful autocorrelation: the Durbin–Watson statistic is 1.99997, essentially equal to 2. Because these are cross-sectional data rather than time-series observations, serial dependence is not the main concern anyway.

index	cooks_distance	leverage	standardized_residual
682	0.035879484	0.054376435	-2.955567557
1698	0.023911381	0.054977132	-2.39881341
957	0.016325061	0.010545981	4.630690343
82	0.014355896	0.005883989	5.827224776
1034	0.01395335	0.034596793	-2.3347464





Lastly, the influence diagnostics exhibits a few, moderately influential points, though none of the observations dominates the model. The maximum Cook distance is 0.0359 which is much less than 1, as observed in the Cook Distance diagram. The Leverage plot indicates that there are a few leverages of between 0.05 and 0.08 and a large number of large standardized residuals, therefore, casewise review would be considerate in a larger analysis. Nevertheless, the data do not indicate a single disastrous outlier that leads to the entire set of findings.

Implemented Remedy: Log Transformation of the Dependent Variable

To address the baseline problems, I implemented a remedial model using a log-transformed dependent variable:

$$log(Y + c)$$

where Y=ViolentCrimesPerPop and c is a small positive shift added because the response includes zero values.

This transformation was chosen for two reasons:

to reduce right-skewness in the response and to stabilize residual variance.

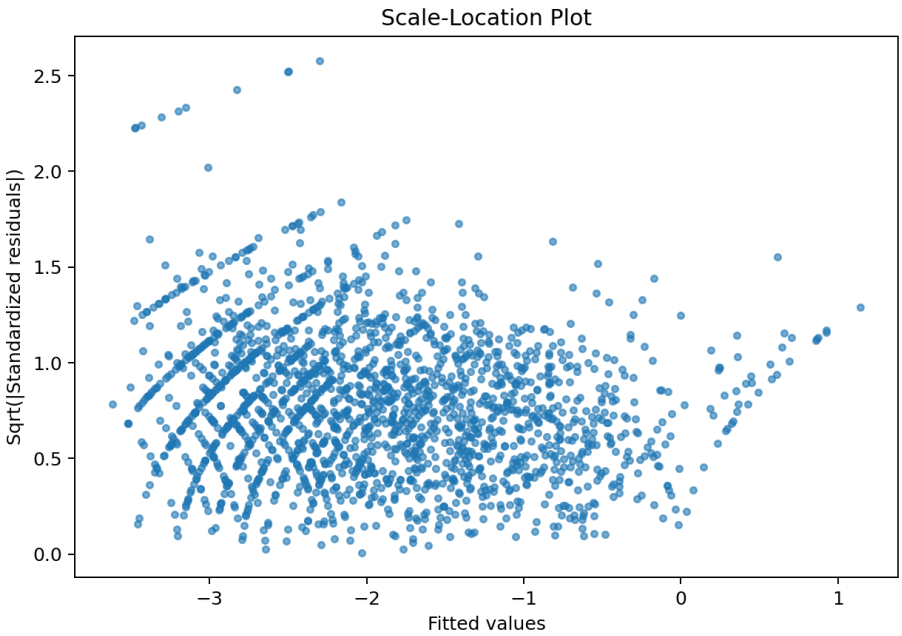
The same 13 predictors were retained in the remedial model so that changes in diagnostics could be attributed to the transformation rather than to a different feature set.

In the baseline model, the response variable is bounded between 0 and 1 and heavily concentrated near lower values, with a long right tail. That kind of outcome often produces asymmetric residuals, fan-shaped variance, and fitted-value-dependent spread.

A log transformation is a standard remedy in such settings because it compresses large outcome values more than small ones. Since the response contains zeros, the shift c makes the transformation well-defined for all observations.

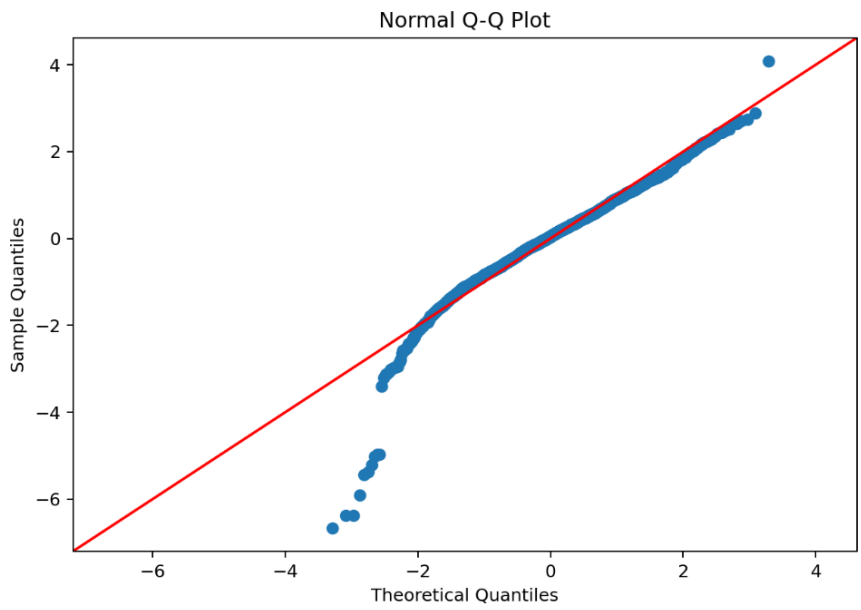
Evaluation of the Remedy

The transformed model was evaluated using the uploaded log-model coefficient table, residual plots, Q–Q plot, leverage/Cook's diagnostics, and influential-point file.



The log-transformed Residuals vs Fitted plot was significantly less dominated by the extreme upper-tail dispersion that was evident in the baseline OLS model. The most evident fan shape was scaled down and the log-model Scale-Location plot was made more even by the middle of the fitted-value range. This implies that the variance was stabilized to some extent by the transformation.

However, the remedial model was not perfect. Owing to the large number of very small or zero values in the response, the transformed residual plots display apparent striping and some residual structure. Thereby, the transformation increased the pattern of variation, but did not entirely remove heteroskedasticity or all the problems with the model forms.



The Q-Q plot that used log-model indicated a significant improvement in the upper tail as compared to the baseline model. In the classical model the right tail was sharply bending over the reference line. The central portion of the distribution was closer to the reference line after transformation and there was less inflation on the upper tail.

At the same time, the lower tail of the transformed model showed a new strong deviation. This is no accident: when a transformation of the log-shift of the response of a bounded response is applied, and the response includes zeros, a very small value will result in a large negative transformed value, and cause left-tail distortion.

So the transformation improved normality partially but not completely:

better in the center and upper tail, worse in the extreme lower tail.

One significant way the log remedy enhanced the influence diagnostics was through its role in enhancing the influence diagnostics. The maximum Cook distance of the baseline was 0.0359, and the maximum Cook distance of the log-model was 0.0284. The maximum leverage remained the same at approximately 0.0791 and it is understandable since leverage is determined by the predictor matrix and the predictors do not vary.

index	cooks_distance	leverage	standardized_residual
342	0.028432895	0.013225601	-5.449740108
773	0.019870089	0.006801257	-6.373638153
1462	0.01673247	0.008543834	-5.213800491
426	0.015429812	0.007444019	-5.366833328
519	0.015138916	0.006056501	-5.897681958

With the typical 4/n rule-of-thumb threshold, the observations above the cutoff of the Cook distance decreased in the transformed model to 78 as compared to 164 in the baseline model. That is, the transformation significantly decreased the number of moderately influential observations, although some large standardized residuals were still present in the low part of the tail.

The remedial model also provides a robustness check for substantive conclusions.

variable	coef	std_err	t	p_value	coef_HC3	std_err_HC3	t_HC3	p_value_HC3	ci_low	ci_high	ci_low_HC3	ci_high_HC3
const	-1.205784785	0.242224844	-4.977956697	6.98E-07	-1.205784785	0.230556053	-5.229898622	1.88E-07	-1.680827143	-0.730742426	-1.657942744	-0.753626825
racepctblack	0.507237536	0.117163262	4.329322425	1.57E-05	0.507237536	0.113799132	4.457305851	8.76E-06	0.277461303	0.737013768	0.284058909	0.730416162
agePct12t29	0.085940603	0.124929017	0.687915466	0.491586595	0.085940603	0.126244413	0.680747776	0.496110697	-0.159065541	0.330946747	-0.161645245	0.333526452
pctUrban	0.279697472	0.038337226	7.295714895	4.28E-13	0.279697472	0.044974382	6.219039828	6.09E-10	0.204511929	0.354883014	0.191495386	0.367899557
OtherPerCap	0.184265096	0.087462354	2.106793229	0.035261012	0.184265096	0.103033082	1.788407097	0.073863199	0.012737178	0.355793013	-0.017799555	0.386329746
MalePctDivorce	1.063912356	0.144681296	7.353489234	2.81E-13	1.063912356	0.136357936	7.802350096	9.76E-15	0.780168776	1.347655936	0.796492241	1.331332471
PctKids2Par	-2.311567081	0.227575497	-10.15736364	1.16E-23	-2.311567081	0.232196254	-9.955229844	8.14E-23	-2.757879685	-1.865254476	-2.766941742	-1.856192419
PctWorkMom	-0.099264148	0.101280487	-0.980091539	0.327160671	-0.099264148	0.111991564	-0.886353795	0.375534547	-0.297891673	0.099363377	-0.318897784	0.120369544
PctIlleg	-0.158902031	0.190531304	-0.833994351	0.404384768	-0.158902031	0.175693478	-0.904427603	0.365878735	-0.532564941	0.214760879	-0.503465549	0.185661486
PctPersDenseHous	1.247153421	0.098664985	12.64028387	2.81E-35	1.247153421	0.091551389	13.62244132	1.85E-40	1.053655325	1.440651522	1.067606241	1.426706061

HousVacant	0.36371746	0.150292651	2.42006152	0.015607383	0.36371746	0.11730292 2	3.100668361	0.001958044	0.06896910 2	0.65846581 9	0.13366733	0.593767591
PctHousOccup	-0.372407047	0.101075331	-3.684450446	0.000235352	-0.372407047	0.09525913	-3.909410545	9.56 E-05	-0.570632228	-0.174181867	-0.559225712	-0.185588383
MedOwnCostPctIncNoMtg	-0.176042367	0.089860589	-1.959060896	0.050245811	-0.176042367	0.084453598	-2.084486286	0.037243835	-0.352273614	0.00018888	-0.341669624	-0.01041511
NumStreet	0.093287055	0.200624621	0.464983082	0.641994702	0.093287055	0.16021141	0.582274723	0.560447944	-0.300170493	0.486744602	-0.220913607	0.407487716

the following predictors remained statistically significant with HC3 robust inference in the log-transformed model:

racepctblack, pctUrban, MalePctDivorce, PctKids2Par, PctPersDenseHous, HousVacant, PctHousOccup, and MedOwnCostPctIncNoMtg.

On the other hand, some predictors that were significant in the baseline model became non-significant or only weakly significant after transformation:

agePct12t29, OtherPerCap, PctWorkMom, PctIlleg, and NumStreet.

This implies that such effects are more responsive to the magnitude of the dependent variable, and thus should be viewed with more reservations. The coefficients in the log model do not have the same interpretation as in the original-scale OLS model. They explain variations in log crime and a shift, rather than variations in the original normalized crime scale.

That is why the transformed model can best be used as a diagnostic treatment and a test of strength, and not as the most intuitive model to display substantive effect sizes.

Conclusion

The Communities and Crime dataset analysis indicates that a 13-variable OLS model can capture a significant portion of the violent crime variation and can be interpreted and statistically robust. The adjusted R^2 of the model was approximately 0.662 whereas the BIC was the lowest in the main OLS candidates, and the predictive performance remained very close to the large models rendering the model the most appropriate explanatory model in this project.

Family structure and housing distress were found to be the most important findings. Family instability was measured by variables like PctKids2Par, PctIlleg, MalePctDivorce, and PctWorkMom, whereas Housing disadvantage and social stress were captured by variables like PctHousOccup, NumStreet, and HousVacant. These patterns were supported by both individual coefficient tests and strong joint F-tests.

Simultaneously, the diagnostic tests indicated that the baseline model had a heteroskedasticity, non-normal residuals, and some misspecification. Two solutions were introduced in order to solve this: HC3 robust standard errors and a log transform of the dependent variable. The log-transformed model better behaved with respect to residual values and decreased the impact of outliers, but did not eliminate all the assumption violations.

As such, the most justifiable ultimate decision would be to retain the original 13-variable OLS model with HC3 strong standard errors as the primary model, and the log-transformed model as a robustness check. In general, the findings indicate that family instability, housing disadvantage, and general socio-economic stress are the best predictors of violent crime in this data, with the available law-enforcement variables being less predictive due to high rates of missingness.